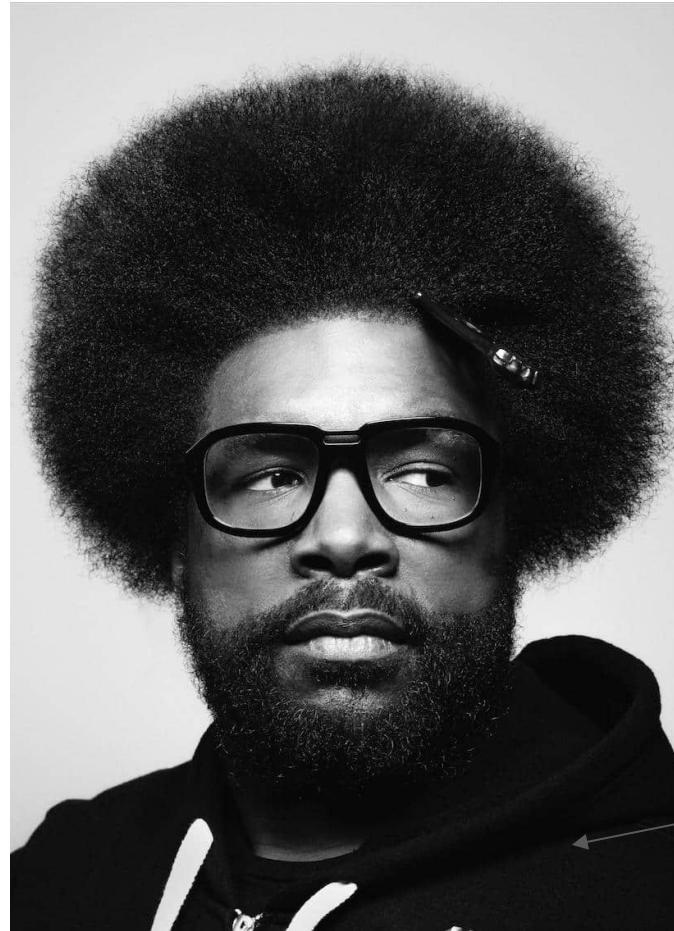




Unit Impulse Superstar

SYDE252 - W05C - Problem Lab

Dr. Matt Borland

Find the natural frequencies and modes of the following systems:

2.2-3

$$(D^3 + 9D)y(t) = (2D^3 + 1)x(t)$$

2.2-4

$$(D^2 + 5D + 6)y(t) = (D + 1)x(t)$$

**Find the unit impulse response
of a system specified by the
equation:**

$$(D^2 + 4D + 3)y(t) = (D + 5)x(t)$$